

MATH 300: MY MATH IDEAS TEXTBOOK (KINOSHITA)

 Σ

Math 300: My Math Ideas Textbook
(Kinoshita)

This text is disseminated via the Open Education Resource (OER) LibreTexts Project (<https://LibreTexts.org>) and like the thousands of other texts available within this powerful platform, it is freely available for reading, printing, and "consuming."

The LibreTexts mission is to bring together students, faculty, and scholars in a collaborative effort to provide an accessible, and comprehensive platform that empowers our community to develop, curate, adapt, and adopt openly licensed resources and technologies; through these efforts we can reduce the financial burden born from traditional educational resource costs, ensuring education is more accessible for students and communities worldwide.

Most, but not all, pages in the library have licenses that may allow individuals to make changes, save, and print this book. Carefully consult the applicable license(s) before pursuing such effects. Instructors can adopt existing LibreTexts texts or Remix them to quickly build course-specific resources to meet the needs of their students. Unlike traditional textbooks, LibreTexts' web based origins allow powerful integration of advanced features and new technologies to support learning.



LibreTexts is the adaptable, user-friendly non-profit open education resource platform that educators trust for creating, customizing, and sharing accessible, interactive textbooks, adaptive homework, and ancillary materials. We collaborate with individuals and organizations to champion open education initiatives, support institutional publishing programs, drive curriculum development projects, and more.

The LibreTexts libraries are Powered by [NICE CXone Expert](#) and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation nor the US Department of Education.

Have questions or comments? For information about adoptions or adaptations contact info@LibreTexts.org or visit our main website at <https://LibreTexts.org>.

The LibreTexts name, logo, and book covers are trademarked by LibreTexts, Inc. (a 501(c)(3) not-for-profit organization) and protected against unauthorized use.

This text was compiled on 08/03/2026

- [InfoPage](#) by [Delmar Larsen](#) is licensed [Public Domain](#).

TABLE OF CONTENTS

Licensing

1: Problem Solving

- 1.1: Problem Solving
 - 1.1.1: Introduction
 - 1.1.2: Percents
 - 1.1.3: Proportions and Rates
 - 1.1.4: Geometry
 - 1.1.5: Problem Solving and Estimating
 - 1.1.6: Exercises

2: Voting Theory and Apportionment

- 2.1: Voting Theory
 - 2.1.1: Introduction
 - 2.1.2: Preference Schedules
 - 2.1.3: Plurality
 - 2.1.4: What's Wrong with Plurality?
 - 2.1.5: Insincere Voting
 - 2.1.6: Instant Runoff Voting
 - 2.1.7: What's Wrong with IRV?
 - 2.1.8: Borda Count
 - 2.1.9: What's Wrong with Borda Count?
 - 2.1.10: Copeland's Method (Pairwise Comparisons)
 - 2.1.11: What's Wrong with Copeland's Method?
 - 2.1.12: So Where's the Fair Method?
 - 2.1.13: Approval Voting
 - 2.1.14: What's Wrong with Approval Voting?
 - 2.1.15: Voting in America
 - 2.1.16: Exercises
 - 2.1.17: Concepts
 - 2.1.18: Exploration
- 2.2: Apportionment
 - 2.2.1: Introduction
 - 2.2.2: Hamilton's Method
 - 2.2.3: Jefferson's Method
 - 2.2.4: Webster's Method
 - 2.2.5: Huntington-Hill Method
 - 2.2.6: Lowndes' Method
 - 2.2.7: Apportionment of Legislative Districts
 - 2.2.8: Exercises

3: Finance

- 3.1: Finance
 - 3.1.1: Introduction
 - 3.1.2: Simple Interest
 - 3.1.3: Compound Interest

- 3.1.4: Annuities
- 3.1.5: Payout Annuities
- 3.1.6: Loans
- 3.1.7: Remaining Loan Balance
- 3.1.8: Which equation to use?
- 3.1.9: Solving for time
- 3.1.10: Exercise

4: Sets

- 4.1: Sets
 - 4.1.1: Basics of Sets
 - 4.1.2: Union, Intersection, and Complement
 - 4.1.3: Venn Diagrams
 - 4.1.4: Cardinality
 - 4.1.5: Exercises

5: Logic

- 5.1: Logic
 - 5.1.1: Logic
 - 5.1.2: Boolean Logic
 - 5.1.3: Conditional Statements
 - 5.1.4: Quantified Statements
 - 5.1.5: Truth Tables- Conjunction (and), Disjunction (or), Negation (not)
 - 5.1.6: Truth Tables- Conditional, Biconditional
 - 5.1.7: Evaluating Deductive Arguments with Truth Tables
 - 5.1.8: Forms of Valid Arguments
 - 5.1.9: Logical Fallacies in Common Language
 - 5.1.10: Exercises
 - 5.1.11: De Morgan's Laws
 - 5.1.12: Arguments
 - 5.1.13: Evaluating Deductive Arguments with Euler Diagrams

[Index](#)

[Glossary](#)

[Detailed Licensing](#)

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Problem Solving

1.1: Problem Solving

1.1.1: Introduction

1.1.2: Percents

1.1.3: Proportions and Rates

1.1.4: Geometry

1.1.5: Problem Solving and Estimating

1.1.6: Exercises

1: Problem Solving is shared under a [not declared](#) license and was authored, remixed, and/or curated by LibreTexts.

SECTION OVERVIEW

1.1: Problem Solving

1.1.1: Introduction

1.1.2: Percents

1.1.3: Proportions and Rates

1.1.4: Geometry

1.1.5: Problem Solving and Estimating

1.1.6: Exercises

Thumbnail: pixabay.com/photos/tax-form-irs-tax-taxes-finance-4080693/

This page titled [1.1: Problem Solving](#) is shared under a [CC BY-SA](#) license and was authored, remixed, and/or curated by [David Lippman](#) ([The OpenTextBookStore](#)) .

1.1.1: Introduction

In previous math courses, you’ve no doubt run into the infamous “word problems.” Unfortunately, these problems rarely resemble the type of problems we actually encounter in everyday life. In math books, you usually are told exactly which formula or procedure to use, and are given exactly the information you need to answer the question. In real life, problem solving requires identifying an appropriate formula or procedure, and determining what information you will need (and won’t need) to answer the question.

In this chapter, we will review several basic but powerful algebraic ideas: percents, rates, and proportions. We will then focus on the problem solving process, and explore how to use these ideas to solve problems where we don’t have perfect information.

This page titled [1.1.1: Introduction](#) is shared under a [CC BY-SA](#) license and was authored, remixed, and/or curated by [David Lippman](#) ([The OpenTextBookStore](#)).

- **1.1: Introduction** by [David Lippman](#) is licensed [CC BY-SA 3.0](#). Original source: <http://www.opentextbookstore.com/mathinsociety>.

1.1.2: Percents

In the 2004 vice-presidential debates, Edwards's claimed that US forces have suffered "90% of the coalition casualties" in Iraq. Cheney disputed this, saying that in fact Iraqi security forces and coalition allies "have taken almost 50 percent" of the casualties[1]. Who is correct? How can we make sense of these numbers?

Percent literally means "per 100," or "parts per hundred." When we write 40%, this is equivalent to the fraction $\frac{40}{100}$ or the decimal 0.40. Notice that 80 out of 200 and 10 out of 25 are also 40%, since $\frac{80}{200} = \frac{10}{25} = \frac{40}{100}$.

✓ Example 1

243 people out of 400 state that they like dogs. What percent is this?

Solution

$$\frac{243}{400} = 0.6075 = \frac{60.75}{100}. \text{ This is } 60.75\%.$$

Notice that the percent can be found from the equivalent decimal by moving the decimal point two places to the right.

✓ Example 2

Write each as a percent: a) $\frac{1}{4}$ b) 0.02 c) 2.35

Solution

$$\text{a) } \frac{1}{4} = 0.25 = 25\% \quad \text{b) } 0.02 = 2\% \quad \text{c) } 2.35 = 235\%$$

📌 Percents

If we have a *part* that is some *percent* of a *whole*, then

$$\text{percent} = \frac{\text{part}}{\text{whole}}, \text{ or equivalently, } \text{part} = \text{percent} \cdot \text{whole}$$

To do the calculations, we write the percent as a decimal.

✓ Example 3

The sales tax in a town is 9.4%. How much tax will you pay on a \$140 purchase?

Solution

Here, \$140 is the whole, and we want to find 9.4% of \$140. We start by writing the percent as a decimal by moving the decimal point two places to the left (which is equivalent to dividing by 100). We can then compute:

$$\text{tax} = 0.094(140) = \$13.16 \text{ in tax.}$$

✓ Example 4

In the news, you hear "tuition is expected to increase by 7% next year." If tuition this year was \$1200 per quarter, what will it be next year?

Solution

The tuition next year will be the current tuition plus an additional 7%, so it will be 107% of this year's tuition:

$$\$1200(1.07) = \$1284.$$

Alternatively, we could have first calculated 7% of \$1200: $\$1200(0.07) = \84 .

Notice this is *not* the expected tuition for next year (we could only wish). Instead, this is the expected *increase*, so to calculate the expected tuition, we'll need to add this change to the previous year's tuition:

$$\$1200 + \$84 = \$1284.$$

? Try it Now 1

A TV originally priced at \$799 is on sale for 30% off. There is then a 9.2% sales tax. Find the price after including the discount and sales tax.

Answer

The sale price is $\$799(0.70) = \559.30 . After tax, the price is $\$559.30(1.092) = \610.76 .

✓ Example 5

The value of a car dropped from \$7400 to \$6800 over the last year. What percent decrease is this?

Solution

To compute the percent change, we first need to find the dollar value change: $\$6800 - \$7400 = -\$600$. Often we will take the absolute value of this amount, which is called the **absolute change**: $|-600| = 600$.

Since we are computing the decrease relative to the starting value, we compute this percent out of \$7400:

$\frac{600}{7400} = 0.081 = 8.1\%$ decrease. This is called a **relative change**.

📌 Absolute and Relative Change

Given two quantities,

Absolute change = |ending quantity - starting quantity|

Relative change: $\frac{\text{absolute change}}{\text{starting quantity}}$

Absolute change has the same units as the original quantity.

Relative change gives a percent change.

The starting quantity is called the **base** of the percent change.

The base of a percent is very important. For example, while Nixon was president, it was argued that marijuana was a “gateway” drug, claiming that 80% of marijuana smokers went on to use harder drugs like cocaine. The problem is, this isn’t true. The true claim is that 80% of harder drug users first smoked marijuana. The difference is one of base: 80% of marijuana smokers using hard drugs, vs. 80% of hard drug users having smoked marijuana. These numbers are not equivalent. As it turns out, only one in 2,400 marijuana users actually go on to use harder drugs[2].

✓ Example 6

There are about 75 QFC supermarkets in the U.S. Albertsons has about 215 stores. Compare the size of the two companies.

Solution

When we make comparisons, we must ask first whether an absolute or relative comparison. The absolute difference is $215 - 75 = 140$. From this, we could say “Albertsons has 140 more stores than QFC.” However, if you wrote this in an article or paper, that number does not mean much. The relative difference may be more meaningful. There are two different relative changes we could calculate, depending on which store we use as the base:

Using QFC as the base, $\frac{140}{75} = 1.867$.

This tells us Albertsons is 186.7% larger than QFC.

Using Albertsons as the base, $\frac{140}{215} = 0.651$.

This tells us QFC is 65.1% smaller than Albertsons.

Notice both of these are showing percent *differences*. We could also calculate the size of Albertsons relative to QFC: , which tells us Albertsons is 2.867 times the size of QFC. Likewise, we could calculate the size of QFC relative to Albertsons: , which tells us

that QFC is 34.9% of the size of Albertsons.

✓ Example 7

Suppose a stock drops in value by 60% one week, then increases in value the next week by 75%. Is the value higher or lower than where it started?

Solution

To answer this question, suppose the value started at \$100. After one week, the value dropped by 60%:

$$\$100 - \$100(0.60) = \$100 - \$60 = \$40.$$

In the next week, notice that base of the percent has changed to the new value, \$40. Computing the 75% increase:

$$\$40 + \$40(0.75) = \$40 + \$30 = \$70.$$

In the end, the stock is still \$30 lower, or $\frac{\$30}{\$100} = 30\%$ lower, valued than it started.

? Try it Now 2

The U.S. federal debt at the end of 2001 was \$5.77 trillion, and grew to \$6.20 trillion by the end of 2002. At the end of 2005 it was \$7.91 trillion, and grew to \$8.45 trillion by the end of 2006[3]. Calculate the absolute and relative increase for 2001-2002 and 2005-2006. Which year saw a larger increase in federal debt?

Answer

2001-2002: Absolute change: \$0.43 trillion. Relative change: 7.45%

2005-2006: Absolute change: \$0.54 trillion. Relative change: 6.83%

2005-2006 saw a larger absolute increase, but a smaller relative increase.

✓ Example 8

A Seattle Times article on high school graduation rates reported “The number of schools graduating 60 percent or fewer students in four years – sometimes referred to as “dropout factories” – decreased by 17 during that time period. The number of kids attending schools with such low graduation rates was cut in half.”

a) Is the “decrease by 17” number a useful comparison?

b) Considering the last sentence, can we conclude that the number of “dropout factories” was originally 34?

Solution

a) This number is hard to evaluate, since we have no basis for judging whether this is a larger or small change. If the number of “dropout factories” dropped from 20 to 3, that’d be a very significant change, but if the number dropped from 217 to 200, that’d be less of an improvement.

b) The last sentence provides relative change which helps put the first sentence in perspective. We can estimate that the number of “dropout factories” was probably previously around 34. However, it’s possible that students simply moved schools rather than the school improving, so that estimate might not be fully accurate.

✓ Example 9

In the 2004 vice-presidential debates, Edwards's claimed that US forces have suffered "90% of the coalition casualties" in Iraq. Cheney disputed this, saying that in fact Iraqi security forces and coalition allies "have taken almost 50 percent" of the casualties. Who is correct?

Solution

Without more information, it is hard for us to judge who is correct, but we can easily conclude that these two percents are talking about different things, so one does not necessarily contradict the other. Edward’s claim was a percent with coalition

forces as the base of the percent, while Cheney's claim was a percent with both coalition and Iraqi security forces as the base of the percent. It turns out both statistics are in fact fairly accurate.

? Try it Now 3

In the 2012 presidential elections, one candidate argued that “the president’s plan will cut \$716 billion from Medicare, leading to fewer services for seniors,” while the other candidate rebuts that “our plan does not cut current spending and actually expands benefits for seniors, while implementing cost saving measures.” Are these claims in conflict, in agreement, or not comparable because they’re talking about different things?

Answer

Without more information, it is hard to judge these arguments. This is compounded by the complexity of Medicare. As it turns out, the \$716 billion is not a cut in current spending, but a cut in future increases in spending, largely reducing future growth in health care payments. In this case, at least the numerical claims in both statements could be considered at least partially true. Here is one source of more information if you’re interested: <http://factcheck.org/2012/08/a-campaign-full-of-medicare/>

We’ll wrap up our review of percents with a couple cautions. First, when talking about a change of quantities that are already measured in percents, we have to be careful in how we describe the change.

✓ Example 10

A politician’s support increases from 40% of voters to 50% of voters. Describe the change.

Solution

We could describe this using an absolute change: $|50\% - 40\%| = 10\%$. Notice that since the original quantities were percents, this change also has the units of percent. In this case, it is best to describe this as an increase of 10 **percentage points**.

In contrast, we could compute the percent change: $\frac{10\%}{40\%} = 0.25 = 25\%$ increase. This is the relative change, and we’d say the politician’s support has increased by 25%.

Lastly, a caution against averaging percents.

✓ Example 11

A basketball player scores on 40% of 2-point field goal attempts, and on 30% of 3-point of field goal attempts. Find the player’s overall field goal percentage.

Solution

It is very tempting to average these values, and claim the overall average is 35%, but this is likely not correct, since most players make many more 2-point attempts than 3-point attempts. We don’t actually have enough information to answer the question. Suppose the player attempted 200 2-point field goals and 100 3-point field goals. Then they made $200(0.40) = 80$ 2-point shots and $100(0.30) = 30$ 3-point shots. Overall, they made 110 shots out of 300, for a $\frac{110}{300} = 0.367 = 36.7\%$ overall field goal percentage.

[1] www.factcheck.org/cheney_edwards_mangle_facts.html

[2] <http://tvtropes.org/pmwiki/pmwiki.php/Main/LiesDamnedLiesAndStatistics>

[3] www.whitehouse.gov/sites/default/files/hist07z1.xls

This page titled [1.1.2: Percents](#) is shared under a [CC BY-SA](#) license and was authored, remixed, and/or curated by [David Lippman](#) ([The OpenTextBookStore](#)).

- **1.2: Percents** by David Lippman is licensed [CC BY-SA 3.0](#). Original source: <http://www.opentextbookstore.com/mathinsociety>.

1.1.3: Proportions and Rates

If you wanted to power the city of Seattle using wind power, how many windmills would you need to install? Questions like these can be answered using rates and proportions.

📌 Rates

A rate is the ratio (fraction) of two quantities.

A **unit rate** is a rate with a denominator of one.

✓ Example 12

Your car can drive 300 miles on a tank of 15 gallons. Express this as a rate.

Solution

Expressed as a rate, $\frac{300 \text{ miles}}{15 \text{ gallons}}$. We can divide to find a unit rate: $\frac{20 \text{ miles}}{1 \text{ gallon}}$, which we could also write as $20 \frac{\text{miles}}{\text{gallon}}$, or just 20 miles per gallon.

📌 Proportion Equation

A proportion equation is an equation showing the equivalence of two rates or ratios.

✓ Example 13

Solve the proportion $\frac{5}{3} = \frac{x}{6}$ for the unknown value x .

Solution

This proportion is asking us to find a fraction with denominator 6 that is equivalent to the fraction $\frac{5}{3}$. We can solve this by multiplying both sides of the equation by 6, giving $x = \frac{5}{3} \cdot 6 = 10$.

✓ Example 14

A map scale indicates that $\frac{1}{2}$ inch on the map corresponds with 3 real miles. How many miles apart are two cities that are $2\frac{1}{4}$ inches apart on the map?

Solution

We can set up a proportion by setting equal two $\frac{\text{map inches}}{\text{real miles}}$ rates, and introducing a variable, x , to represent the unknown quantity – the mile distance between the cities.

$$\frac{\frac{1}{2} \text{ map inch}}{3 \text{ miles}} = \frac{2\frac{1}{4} \text{ map inches}}{x \text{ miles}} \quad \text{Multiply both sides by } x \text{ and rewriting the mixed number}$$

$$\frac{1}{3} \cdot x = \frac{9}{4} \quad \text{Multiply both sides by 3}$$

$$\frac{1}{2}x = \frac{27}{4} \quad \text{Multiply both sides by 2 (or divide by } \frac{1}{2})$$

$$x = \frac{27}{2} = 13\frac{1}{2} \text{ miles}$$

Many proportion problems can also be solved using **dimensional analysis**, the process of multiplying a quantity by rates to change the units.

✓ Example 15

Your car can drive 300 miles on a tank of 15 gallons. How far can it drive on 40 gallons?

Solution

We could certainly answer this question using a proportion: $\$ \frac{300 \text{ miles}}{15 \text{ gallons}} = \frac{x \text{ miles}}{40 \text{ gallons}} \$$.

However, we earlier found that 300 miles on 15 gallons gives a rate of 20 miles per gallon. If we multiply the given 40 gallon quantity by this rate, the *gallons* unit “cancels” and we’re left with a number of miles:

$$40 \text{ gallons} \cdot \frac{20 \text{ miles}}{1 \text{ gallon}} = \frac{40 \text{ gallons}}{1} \cdot \frac{20 \text{ miles}}{\text{gallon}} = 800 \text{ miles}$$

Notice if instead we were asked “how many gallons are needed to drive 50 miles?” we could answer this question by inverting the 20 mile per gallon rate so that the *miles* unit cancels and we’re left with gallons:

$$50 \text{ miles} \cdot \frac{1 \text{ gallon}}{20 \text{ miles}} = \frac{50 \text{ miles}}{1} \cdot \frac{1 \text{ gallon}}{20 \text{ miles}} = \frac{50 \text{ gallons}}{20} = 2.5 \text{ gallons}$$

Dimensional analysis can also be used to do unit conversions. Here are some unit conversions for reference.

Unit Conversions

Length

1 foot (ft) = 12 inches (in) 1 yard (yd) = 3 feet (ft)
 1 mile = 5,280 feet
 1000 millimeters *mm* = 1 meter (m) 100 centimeters (cm) = 1 meter
 1000 meters (m) = 1 kilometer (km) 2.54 centimeters (cm) = 1 inch

Weight and Mass

1 pound (lb) = 16 ounces (oz) 1 ton = 2000 pounds
 1000 milligrams (mg) = 1 gram (g) 1000 grams = 1 kilogram (kg)
 1 kilogram = 2.2 pounds (on earth)

Capacity

1 cup = 8 fluid ounces (fl oz)* 1 pint = 2 cups
 1 quart = 2 pints = 4 cups 1 gallon = 4 quarts = 16 cups
 1000 milliliters (ml) = 1 liter (L)

*Fluid ounces are a capacity measurement for liquids. 1 fluid ounce $\approx$ 1 ounce (weight) for water only.

Example 16

A bicycle is traveling at 15 miles per hour. How many feet will it cover in 20 seconds?

Solution

To answer this question, we need to convert 20 seconds into feet. If we know the speed of the bicycle in feet per second, this question would be simpler. Since we don’t, we will need to do additional unit conversions. We will need to know that 5280 ft = 1 mile. We might start by converting the 20 seconds into hours:

$$\begin{aligned} 20 \text{ seconds} &\cdot \frac{1 \text{ minute}}{60 \text{ seconds}} \cdot \frac{1 \text{ hour}}{60 \text{ minutes}} = \frac{1}{180} \text{ hour} && \text{Now we can multiply by the 15 miles/hr} \\ \frac{1}{180} \text{ hour} &\cdot \frac{15 \text{ miles}}{1 \text{ hour}} = \frac{1}{12} \text{ mile} && \text{Now we can convert to feet} \\ \frac{1}{12} \text{ mile} &\cdot \frac{5280 \text{ feet}}{1 \text{ mile}} = 440 \text{ feet} \end{aligned}$$

We could have also done this entire calculation in one long set of products:

$$20 \text{ seconds} \cdot \frac{1 \text{ minute}}{60 \text{ seconds}} \cdot \frac{1 \text{ hour}}{60 \text{ minutes}} \cdot \frac{15 \text{ miles}}{1 \text{ hour}} \cdot \frac{5280 \text{ feet}}{1 \text{ mile}} = 440 \text{ feet}$$

? Try it Now 4

A 1000 foot spool of bare 12-gauge copper wire weighs 19.8 pounds. How much will 18 inches of the wire weigh, in ounces?

Answer

$$18 \text{ inches} \cdot \frac{1 \text{ foot}}{12 \text{ inches}} \cdot \frac{19.8 \text{ pounds}}{1000 \text{ feet}} \cdot \frac{16 \text{ ounces}}{1 \text{ pound}} \approx 0.475 \text{ ounces}$$

Notice that with the miles per gallon example, if we double the miles driven, we double the gas used. Likewise, with the map distance example, if the map distance doubles, the real-life distance doubles. This is a key feature of proportional relationships, and

one we must confirm before assuming two things are related proportionally.

✓ Example 17

Suppose you're tiling the floor of a 10 ft by 10 ft room, and find that 100 tiles will be needed. How many tiles will be needed to tile the floor of a 20 ft by 20 ft room?

Solution

In this case, while the width the room has doubled, the area has quadrupled. Since the number of tiles needed corresponds with the area of the floor, not the width, 400 tiles will be needed. We could find this using a proportion based on the areas of the rooms:

$$\frac{100 \text{ tiles}}{100\text{ft}^2} = \frac{n \text{ tiles}}{400\text{ft}^2}$$

Other quantities just don't scale proportionally at all.

✓ Example 18

Suppose a small company spends \$1000 on an advertising campaign, and gains 100 new customers from it. How many new customers should they expect if they spend \$10,000?

Solution

While it is tempting to say that they will gain 1000 new customers, it is likely that additional advertising will be less effective than the initial advertising. For example, if the company is a hot tub store, there are likely only a fixed number of people interested in buying a hot tub, so there might not even be 1000 people in the town who would be potential customers.

Sometimes when working with rates, proportions, and percents, the process can be made more challenging by the magnitude of the numbers involved. Sometimes, large numbers are just difficult to comprehend.

✓ Example 19

Compare the 2010 U.S. military budget of \$683.7 billion to other quantities.

Here we have a very large number, about \$683,700,000,000 written out. Of course, imagining a billion dollars is very difficult, so it can help to compare it to other quantities.

If that amount of money was used to pay the salaries of the 1.4 million Walmart employees in the U.S., each would earn over \$488,000.

There are about 300 million people in the U.S. The military budget is about \$2,200 per person.

If you were to put \$683.7 billion in \$100 bills, and count out 1 per second, it would take 216 years to finish counting it.

✓ Example 20

Compare the electricity consumption per capita in China to the rate in Japan.

To address this question, we will first need data. From the CIA[1] website we can find the electricity consumption in 2011 for China was 4,693,000,000,000 KWH (kilowatt-hours), or 4.693 trillion KWH, while the consumption for Japan was 859,700,000,000, or 859.7 billion KWH. To find the rate per capita (per person), we will also need the population of the two countries. From the World Bank[2], we can find the population of China is 1,344,130,000, or 1.344 billion, and the population of Japan is 127,817,277, or 127.8 million.

Solution

Computing the consumption per capita for each country:

$$\text{China: } \frac{4,693,000,000,000 \text{ KWH}}{1,344,130,000 \text{ people}} \approx 3491.5 \text{ KWH per person}$$

$$\text{Japan: } \frac{859,700,000,000 \text{ KWH}}{127,817,277 \text{ people}} \approx 6726 \text{ KWH per person}$$

While China uses more than 5 times the electricity of Japan overall, because the population of Japan is so much smaller, it turns out Japan uses almost twice the electricity per person compared to China.

[1] www.cia.gov/library/publicat.../2042rank.html

[2] <http://data.worldbank.org/indicator/SP.POP.TOT>

This page titled [1.1.3: Proportions and Rates](#) is shared under a [CC BY-SA](#) license and was authored, remixed, and/or curated by [David Lippman](#) ([The OpenTextBookStore](#)) .

- **1.3: Proportions and Rates** by [David Lippman](#) is licensed [CC BY-SA 3.0](#). Original source: <http://www.opentextbookstore.com/mathinsociety>.

1.1.4: Geometry

Geometric shapes, as well as area and volumes, can often be important in problem solving.

✓ Example 21

You are curious how tall a tree is, but don't have any way to climb it. Describe a method for determining the height.

There are several approaches we could take. We'll use one based on triangles, which requires that it's a sunny day. Suppose the tree is casting a shadow, say 15 ft long. I can then have a friend help me measure my own shadow. Suppose I am 6 ft tall, and cast a 1.5 ft shadow. Since the triangle formed by the tree and its shadow has the same angles as the triangle formed by me and my shadow, these triangles are called **similar triangles** and their sides will scale proportionally. In other words, the ratio of height to width will be the same in both triangles. Using this, we can find the height of the tree, which we'll denote by h :

Solution

$$\frac{6\text{ft tall}}{1.5\text{ft shadow}} = \frac{h\text{ft tall}}{15\text{ft shadow}}$$

Multiplying both sides by 15, we get $h = 60$. The tree is about 60 ft tall.

It may be helpful to recall some formulas for areas and volumes of a few basic shapes.

📌 Areas

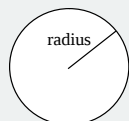
Rectangle



Area: $L \cdot W$

Perimeter: $2L + 2W$

Circle, radius r

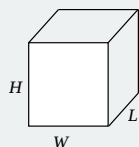


Area: πr^2

Circumference = $2\pi r$

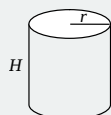
📌 Volumes

Rectangular Box



Volume: $L \cdot W \cdot H$

Cylinder



✓ Example 22

If a 12 inch diameter pizza requires 10 ounces of dough, how much dough is needed for a 16 inch pizza?

Solution

To answer this question, we need to consider how the weight of the dough will scale. The weight will be based on the volume of the dough. However, since both pizzas will be about the same thickness, the weight will scale with the area of the top of the pizza. We can find the area of each pizza using the formula for area of a circle, $A = \pi r^2$:

A 12" pizza has radius 6 inches, so the area will be $\pi 6^2 =$ about 113 square inches.

A 16" pizza has radius 8 inches, so the area will be $\pi 8^2 =$ about 201 square inches.

Notice that if both pizzas were 1 inch thick, the volumes would be 113 in^3 and 201 in^3 respectively, which are at the same ratio as the areas. As mentioned earlier, since the thickness is the same for both pizzas, we can safely ignore it.

We can now set up a proportion to find the weight of the dough for a 16" pizza:

$$\frac{10 \text{ ounces}}{113 \text{ in}^2} = \frac{x \text{ ounces}}{201 \text{ in}^2} \quad \text{Multiply both sides by 201}$$

$$x = 201 \cdot \frac{10}{113} = \text{about } 17.8 \text{ ounces of dough for a 16" pizza.}$$

It is interesting to note that while the diameter is $\frac{16}{12} = 1.33$ times larger, the dough required, which scales with area, is $1.33^2 = 1.78$ times larger.

✓ Example 23

A company makes regular and jumbo marshmallows. The regular marshmallow has 25 calories. How many calories will the jumbo marshmallow have?

We would expect the calories to scale with volume. Since the marshmallows have cylindrical shapes, we can use that formula to find the volume. From the grid in the image, we can estimate the radius and height of each marshmallow.



Solution

The regular marshmallow appears to have a diameter of about 3.5 units, giving a radius of 1.75 units, and a height of about 3.5 units. The volume is about $\pi(1.75)^2(3.5) = 33.7 \text{ units}^3$.

The jumbo marshmallow appears to have a diameter of about 5.5 units, giving a radius of 2.75 units, and a height of about 5 units. The volume is about $\pi(2.75)^2(5) = 118.8 \text{ units}^3$.

We could now set up a proportion, or use rates. The regular marshmallow has 25 calories for 33.7 cubic units of volume. The jumbo marshmallow will have:

$$118.8 \text{ units}^3 \cdot \frac{25 \text{ calories}}{33.7 \text{ units}^3} = 88.1 \text{ calories}$$

It is interesting to note that while the diameter and height are about 1.5 times larger for the jumbo marshmallow, the volume and calories are about $1.5^3 = 3.375$ times larger.

? Try it Now 5

A website says that you'll need 48 fifty-pound bags of sand to fill a sandbox that measure 8ft by 8ft by 1ft. How many bags would you need for a sandbox 6ft by 4ft by 1ft?

Answer

The original sandbox has volume 64 ft^3 . The smaller sandbox has volume 24 ft^3 .

$$\frac{48 \text{ bags}}{64 \text{ ft}^3} = \frac{x \text{ bags}}{24 \text{ ft}^3} \quad \text{results in } x = 18 \text{ bags.}$$

This page titled [1.1.4: Geometry](#) is shared under a [CC BY-SA](#) license and was authored, remixed, and/or curated by [David Lippman](#) ([The OpenTextBookStore](#)) .

- [1.4: Geometry](#) by [David Lippman](#) is licensed [CC BY-SA 3.0](#). Original source: <http://www.opentextbookstore.com/mathinsociety>.